



**Final**  
**MISO DPP 2019 Cycle 1 South Area**  
**Study Phase III Report**

November 18, 2021

**MISO**  
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## 1. Executive Summary

This report presents the results of a System Impact Study performed to evaluate the interconnection of the generators in the DPP 2019 Cycle 1 South Area Phase III. The study was performed under the direction of MISO and reviewed by an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Entergy, Cooperative Energy, CLECO, AECC, AECL, Southern Company, PowerSouth, SPP, and TVA.

### 1.1. Project List

The original interconnection requests for DPP 2019 Cycle 1 South Area had a total of 70 generation projects. Projects J1283, J1291, J1309, J1433, J1443, and J1484 withdrew prior to the start of DPP Phase I. Projects J1220, J1236, J1238, J1245, J1250, J1259, J1260, J1261, J1267, J1277, J1278, J1280, J1287, J1290, J1293, J1351, J1364, J1380, J1384, J1400, J1404, J1414, J1439, J1462, and J1469 withdrew during Decision Point I prior to the start of DPP Phase II. Projects J1217, J1223, J1279, J1282, and J1452 withdrew during Decision Point II. Therefore, there are 34 generation projects with a combined nameplate rating of 5305.18 MW of Energy Resource Interconnection Service (ERIS) & 4077.6 MW of Network Resource Interconnection Service (NRIS). The detailed list of South Area DPP 1 is shown below in Table 1, and the Phase III study was kicked off on March 22<sup>nd</sup>, 2021.

**Table 1: List of DPP 2019 Cycle 1 South Area Phase II Projects**

| Project | Fuel Type           | Transmission Owner  | County                          | State | Service Requested | MW (ERIS/NRIS If different) | Point of Interconnection (POI)       |
|---------|---------------------|---------------------|---------------------------------|-------|-------------------|-----------------------------|--------------------------------------|
| J1205   | Solar               | Cleco Corporation   | Allen Parish, Assumption Parish | LA    | NRIS              | 125                         | Centennial - West Bay 138 kV         |
| J1215   | Solar               | Entergy Arkansas    | Arkansas County                 | AR    | NRIS              | 155                         | Stuttgart Rikuskey – Woodward 230 kV |
| J1219   | Solar               | Entergy Louisiana   | Tangipahoa Parish               | LA    | NRIS              | 100                         | Chicken Farm – Amite 115 kV          |
| J1239   | Solar               | Entergy Louisiana   | Ouachita Parish                 | LA    | NRIS              | 100/0                       | Rilla 115 kV Substation              |
| J1246   | Gas                 | Entergy Louisiana   | Iberville Parish                | LA    | NRIS              | 705.2                       | Bayou Labutte 500 kV Substation      |
| J1247   | Waste Heat Recovery | Cleco Corporation   | St Mary Parish                  | LA    | ERIS              | 30.98/0                     | Ivanhoe 138 kV Substation            |
| J1249   | Gas                 | Cooperative Energy  | Lamar County                    | MS    | NRIS              | 600                         | Morrow 161 kV Substation             |
| J1257   | Solar               | Entergy Louisiana   | Tangipahoa County               | LA    | NRIS              | 20                          | Chicken Farm – Amite 115 kV          |
| J1258   | Solar               | Entergy Arkansas    | Lonohe County                   | AR    | NRIS              | 50                          | KEO EHV 500 kV Substation            |
| J1281   | Solar               | Entergy Arkansas    | Ashley County                   | AR    | NRIS              | 75                          | Crossett – Monticello East 115 kV    |
| J1286   | Solar               | Cleco Corporation   | Evangeline Parish               | LA    | NRIS              | 125/0                       | Manuel 138 kV Substation             |
| J1294   | Solar               | Entergy Mississippi | Sunflower County                | MS    | NRIS              | 99                          | Indianola – Cleveland 115 kV         |
| J1347   | Solar               | Entergy Mississippi | Scott County                    | MS    | ERIS              | 175/0                       | Pelahatchie – Scott 115 kV           |
| J1361   | Solar               | Entergy             | Washington                      | LA    | NRIS              | 50                          | Lagen Holton – Franklinton 115 kV    |

| Project | Fuel Type       | Transmission Owner                        | County                  | State | Service Requested | MW (ERIS/NRIS If different) | Point of Interconnection (POI)                   |
|---------|-----------------|-------------------------------------------|-------------------------|-------|-------------------|-----------------------------|--------------------------------------------------|
|         |                 | Louisiana                                 | Parish                  |       |                   |                             |                                                  |
| J1367   | Solar           | Cleco Corporation                         | St. Mary Parish         | LA    | NRIS              | 95/48.65                    | Caneland 230 kV Substation                       |
| J1368   | Solar           | Entergy Louisiana                         | Point Coupee Parish     | LA    | NRIS              | 200/40                      | Fancy Point – Waterloo 230 kV                    |
| J1372   | Solar           | Entergy Louisiana                         | Point Coupee Parish     | LA    | NRIS              | 100/20                      | Fancy Point – Waterloo 230 kV                    |
| J1373   | Solar           | Entergy Arkansas                          | White County            | AR    | NRIS              | 95/80.75                    | Griffithville – Searcy Price 115 kV              |
| J1402   | Battery Storage | Entergy Arkansas                          | Jackson County          | AR    | NA                | 0                           | Cash – Newport AB 161 kV                         |
| J1415   | Solar           | Entergy Arkansas                          | Lonoque County          | AR    | NRIS              | 250                         | KEO EHV 500 kV Substation                        |
| J1421   | Solar           | Entergy Louisiana                         | East Feliciana Parish   | LA    | NRIS              | 300                         | Moler - Colfel 230 kV                            |
| J1424   | Solar           | Cleco Corporation                         | Beauregard Parish       | LA    | NRIS              | 200/0                       | Cooper – Penton 230 kV                           |
| J1425   | Gas             | Entergy Arkansas                          | Jefferson County        | AR    | NRIS              | 0/114                       | Pine Bluff International Paper 115 kV Substation |
| J1431   | Solar           | Entergy Louisiana                         | St. James Parish        | LA    | NRIS              | 90                          | Vacherie 230 kV Substation                       |
| J1434   | Solar           | Entergy Arkansas                          | Monroe County           | AR    | NRIS              | 100                         | Brinkley East – Aubrey 230 kV                    |
| J1436   | Solar           | Entergy Louisiana                         | Morehouse Parish        | LA    | ERIS              | 50                          | Oak Ridge 115 kV Substation                      |
| J1437   | Wind            | Arkansas Electric Cooperative Corporation | Carroll County          | AR    | NRIS              | 180                         | AECC Dry Creek 161 kV Switchyard Station         |
| J1440   | Solar           | Entergy Mississippi                       | Tunica County           | MS    | NRIS              | 150                         | Lula – Tunica 115 kV                             |
| J1441   | Solar           | Entergy Louisiana                         | Terrebonne Parish       | LA    | NRIS              | 200                         | Schriever 230 kV Substation                      |
| J1442   | Solar           | Entergy Texas                             | Polk County             | TX    | NRIS              | 105/0                       | Rich (SHECO) – Livingston 138 kV                 |
| J1455   | Solar           | Cleco Corporation                         | St. Mary Parish         | LA    | NRIS              | 150                         | Teche – Morbihan 138 kV                          |
| J1458   | Solar           | Entergy Mississippi                       | Washington County       | MS    | NRIS              | 200/170                     | Ray Braswell – Tinnin – Gerald Andrus 230 kV     |
| J1463   | Solar           | Entergy Louisiana                         | West Baton Rouge Parish | LA    | NRIS              | 150/0                       | Addis – Big Cajun 230 kV                         |
| J1465   | Solar           | Entergy Louisiana                         | Pointe Coupee Parish    | LA    | NRIS              | 150                         | Livonia – Colonial Springs 138 kV                |

## 1.2. Total Network Upgrades

The cost allocation of Network Upgrades for the projects in the DPP 2019 Cycle 1 South Phase III reflects responsibilities for mitigating system impacts. The total cost of Network Upgrades is listed in Table 2 below. The costs for Network Upgrades are planning-level estimates and subject to revision in the facility studies. Details pertaining to the cost allocation has been detailed in Appendix A – Cost Allocation Summary (CEII).

**Table 2: Total Cost of Network Upgrades for DPP 2019 Cycle 1 South Phase III Projects**

| Project | ERIS Network Upgrades (\$) |           |               |                 | NRIS Network Upgrades (\$) | Interconnection Facilities (\$) |                            | Shared Network Upgrades (\$) | Total Network Upgrade Cost for Milestone Calculation (\$) | M2 (\$)    | M3 (\$)                     | M4 (\$)                              |
|---------|----------------------------|-----------|---------------|-----------------|----------------------------|---------------------------------|----------------------------|------------------------------|-----------------------------------------------------------|------------|-----------------------------|--------------------------------------|
|         | Thermal                    | Stability | Short Circuit | Affected System | Deliverability             | TO Network Upgrades             | TO – Owned Direct Assigned |                              |                                                           |            |                             |                                      |
| a       | b                          | c         | d             | e               | f                          | g                               | h                          | i                            | j = b+c+d+f+g+i                                           | \$4,000/MW | 10% of (j) from Phase I- M2 | (20% of (j) from Phase II) – M2 – M3 |
| J1205   | -                          | -         | -             | -               | -                          | 10,094,000                      | 1,075,000                  | -                            | 10,094,000                                                | 500,000    | 3,475,319                   | 4,786,751                            |
| J1215   | -                          | -         | -             | -               | 341,594                    | 13,149,000                      | 1,546,000                  | -                            | 13,490,594                                                | 800,000    | 4,252,878                   | -                                    |
| J1219   | -                          | -         | -             | -               | -                          | 5,946,000                       | 1,249,000                  | -                            | 5,946,000                                                 | 400,000    | 350,977                     | 438,223                              |
| J1239   | -                          | -         | -             | -               | -                          | 3,802,000                       | 3,431,000                  | -                            | 3,802,000                                                 | 400,000    | 4,016,325                   | -                                    |
| J1246   | -                          | -         | 30,613,760    | -               | 60,484,117                 | 10,648,000                      | 2,524,000                  | -                            | 101,745,877                                               | 2,900,800  | 3,378,495                   | 4,631,709                            |
| J1247   | -                          | -         | -             | -               | -                          | -                               | 2,928,018                  | -                            | -                                                         | 124,000    | -                           | -                                    |
| J1249   | 6,772,000                  | -         | -             | 63,300,000      | -                          | 7,550,000                       | 1,160,000                  | -                            | 14,322,000                                                | 2,400,000  | -                           | 464,400                              |
| J1257   | -                          | -         | -             | -               | -                          | 5,946,000                       | 1,249,000                  | -                            | 5,946,000                                                 | 80,000     | 670,195                     | 439,005                              |
| J1258   | -                          | -         | -             | -               | -                          | 1,959,500                       | 2,484,500                  | -                            | 1,959,500                                                 | 200,000    | 282,027                     | -                                    |
| J1281   | -                          | -         | -             | -               | 2,742,182                  | 11,779,000                      | 1,507,000                  | -                            | 14,521,182                                                | 300,000    | 1,828,655                   | 693,401                              |
| J1286   | -                          | -         | -             | -               | -                          | 2,426,089                       | 195,433                    | -                            | 2,426,089                                                 | 600,000    | 5,978,388                   | 7,612,356                            |
| J1294   | -                          | -         | -             | -               | 4,871,549                  | 12,015,000                      | 1,517,000                  | -                            | 16,886,549                                                | 396,000    | 1,831,523                   | 1,013,299                            |
| J1347   | -                          | -         | -             | 17,000,000      | -                          | 12,910,000                      | 1,515,000                  | -                            | 12,910,000                                                | 700,000    | 1,822,322                   | 59,678                               |
| J1361   | -                          | -         | -             | -               | -                          | 1,364,000                       | -                          | -                            | 1,364,000                                                 | 200,000    | 200,480                     | -                                    |
| J1367   | -                          | -         | -             | -               | 384,940                    | 3,650,000                       | 1,503,000                  | -                            | 4,034,940                                                 | 380,000    | 923,623                     | 889,330                              |
| J1368   | -                          | -         | -             | -               | -                          | 2,605,000                       | 72,000                     | -                            | 2,605,000                                                 | 800,000    | 5,418,271                   | 3,821,771                            |
| J1372   | -                          | -         | -             | -               | -                          | 2,605,000                       | 72,000                     | -                            | 2,605,000                                                 | 400,000    | 3,084,135                   | 2,285,885                            |
| J1373   | -                          | -         | -             | -               | -                          | 3,588,000                       | 129,000                    | -                            | 3,588,000                                                 | 380,000    | 2,117,528                   | -                                    |
| J1402   | -                          | -         | -             | -               | -                          | -                               | -                          | -                            | -                                                         | -          | -                           | -                                    |
| J1415   | -                          | -         | -             | -               | -                          | 1,959,500                       | 2,484,500                  | -                            | 1,959,500                                                 | 1,000,000  | -                           | -                                    |
| J1421   | 19,600,000                 | -         | 955,035       | -               | -                          | 21,305,000                      | 95,000                     | -                            | 41,860,035                                                | 1,200,000  | 1,553,225                   | 4,166,775                            |
| J1424   | -                          | -         | -             | -               | -                          | 8,978,660                       | 322,040                    | -                            | 8,978,660                                                 | 800,000    | 8,439,655                   | 11,179,762                           |

| Project | ERIS Network Upgrades (\$) |           |               |                 | NRIS Network Upgrades (\$) | Interconnection Facilities (\$) |                            | Shared Network Upgrades (\$) | Total Network Upgrade Cost for Milestone Calculation (\$) | M2 (\$)    | M3 (\$)                     | M4 (\$)                              |
|---------|----------------------------|-----------|---------------|-----------------|----------------------------|---------------------------------|----------------------------|------------------------------|-----------------------------------------------------------|------------|-----------------------------|--------------------------------------|
|         | Thermal                    | Stability | Short Circuit | Affected System | Deliverability             | TO Network Upgrades             | TO – Owned Direct Assigned |                              |                                                           |            |                             |                                      |
| a       | b                          | c         | d             | e               | f                          | g                               | h                          | i                            | j = b+c+d+f+g+i                                           | \$4,000/MW | 10% of (j) from Phase I- M2 | (20% of (j) from Phase II) – M2 – M3 |
| J1425   | -                          | -         | -             | -               | 232,501                    | -                               | -                          | -                            | 232,501                                                   | 840,000    | -                           | -                                    |
| J1431   | -                          | -         | -             | -               | 7,477,678                  | 608,000                         | 608,000                    | -                            | 8,085,678                                                 | 800,000    | 3,382,034                   | -                                    |
| J1434   | -                          | -         | -             | -               | 227,480                    | -                               | -                          | -                            | 227,480                                                   | 400,000    | 1,921,525                   | -                                    |
| J1436   | -                          | -         | -             | -               | -                          | -                               | -                          | -                            | -                                                         | 200,000    | 157,619                     | -                                    |
| J1437   | -                          | -         | -             | 1,150,239       | -                          | 3,418,000                       | 1,090,000                  | -                            | 3,418,000                                                 | 720,000    | -                           | -                                    |
| J1440   | -                          | -         | -             | -               | 407,336                    | 12,064,000                      | 1,496,000                  | -                            | 12,471,336                                                | 640,000    | 2,419,980                   | -                                    |
| J1441   | -                          | -         | -             | -               | 16,822,322                 | 3,255,000                       | 3,937,000                  | -                            | 20,077,322                                                | 800,000    | 3,675,142                   | 100,170                              |
| J1442   | -                          | -         | -             | -               | -                          | 13,578,000                      | 2,012,000                  | -                            | 13,578,000                                                | 660,000    | 5,668,331                   | 5,244,093                            |
| J1455   | -                          | -         | -             | -               | 1,267,830                  | 10,814,000                      | 1,071,000                  | -                            | 12,081,830                                                | 600,000    | 1,957,465                   | 2,012,035                            |
| J1458   | -                          | -         | -             | -               | 14,977,358                 | 4,380,000                       | 136,000                    | -                            | 19,357,358                                                | 800,000    | 3,839,272                   | -                                    |
| J1463   | -                          | -         | 431,347       | -               | -                          | -                               | 128,000                    | -                            | 431,347                                                   | 600,000    | 2,934,367                   | 951,087                              |
| J1465   | -                          | -         | -             | -               | 363,113                    | 14,913,000                      | 2,820,000                  | -                            | 15,276,113                                                | 600,000    | 3,130,304                   | 5,768,361                            |

Analyses performed demonstrate the following transmission facilities are required to reliably interconnect this group of generators to the transmission system. ERIS Network Upgrades and NRIS Network Upgrades are shown in Table 3.

**Table 3: ERIS & NRIS Upgrades (Planning level cost estimates)**

| Network Upgrade                                                      | TO  | GI projects requiring upgrade for ERIS | GI projects requiring upgrade for NRIS          | Cost of solution (\$) | TO Self-Fund Election |
|----------------------------------------------------------------------|-----|----------------------------------------|-------------------------------------------------|-----------------------|-----------------------|
| Morrow - Purvis Bulk 161 kV New Line, Upgrade Purvis Bulk Substation | CE  | J1249                                  |                                                 | \$ 6,722,000          | Yes                   |
| Bayou Labutte 500 kV Expansion + New Auto + New Line to Wise         | EES |                                        | J1246, J1367, J1455, J1465                      | \$ 62,500,000         | No                    |
| Bogalusa - Adams Creek 230 kV Upgrade                                | EES |                                        | J1281, J1294, J1458                             | \$ 15,000,000         | No                    |
| Coly - Moler 230 kV Upgrade                                          | EES | J1421                                  |                                                 | \$ 19,600,000         | No                    |
| Greenville - SE Greenville 115 kV Reconductor                        | EES |                                        | J1458                                           | \$ 5,300,000          | No                    |
| Isola SS - Belzoni 115 kV Upgrade                                    | EES |                                        | J1294, J1458                                    | \$ 500,000            | No                    |
| Gypsy - Ironman 230 kV Upgrade                                       | EES |                                        | J1431, J1441                                    | \$ 500,000            | No                    |
| Frisco - Convent 230 kV Upgrade                                      | EES |                                        | J1431, J1441                                    | \$ 23,800,000         | No                    |
| Ray Braswell - Franklin 500 kV Upgrade                               | EES |                                        | J1215, J1281, J1294, J1425, J1434, J1440, J1458 | \$ 3,000,000          | No                    |
| Willow Glen 230 kV Breaker Replacements                              | EES |                                        | J1246, J1421, J1463                             | \$ 28,500,000         | No                    |
| Pt. Pleasant 230 kV Breaker Replacements                             | EES |                                        | J1246, J1421, J1463                             | \$ 3,500,000          | No                    |

**Note:**

- 1) Details pertaining to upgrades, costs, and the execution plan for interconnection of the generating facility at the POI will be documented in the Facility Study for Interconnecting Generator.
- 2) Facilities that have been included as base case assumptions and the level of interconnection service that would be conditional upon these facilities being in service will be documented in the GIA (Generator Interconnection Agreement) for each respective GI request successfully achieving GIA execution.
- 3) A detailed summary of the estimated cost of Network Upgrades on a per project basis can be found in Appendix B – Network Upgrade Table Per Project (CEII).

## 2. FERC Order 827 Compliance Review

The Final Rule of FERC Order 827 “Reactive Power Requirements for Non-Synchronous Generation”, which was issued June 16, 2016, states that “Under this Final Rule, newly interconnecting non-synchronous generators that have not yet executed a Facilities Study Agreement as of the effective date of this Final Rule will be required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.” As such, this Final Rule applies to all wind and solar projects included in the DPP 2019 Cycle 1 South study cycle.

In this study, the power factor at the high-side of the generator substation for each inverter-based project was calculated and reviewed. The study method is to set Qgen of each study project at its Qmax, solve the case, then record the P and Q injection on the high side of the generator substation to calculate the lagging power factor (injecting VAR to the system). The same process is then repeated by setting Qgen at Qmin to calculate the leading power factor (absorbing VAR from the system).

The results show that all projects meet the requirement to maintain 0.95 leading power factor, however, 32 projects do not meet the requirement to provide reactive power capability corresponding to 0.95 lagging power factor, as highlighted in red below in Table 4. Additional reactive support will be needed for these projects to meet the FERC requirement on reactive power capability prior to the completion of their GIA.

**Table 4: FERC Order 827 Review Results**

| Project        | Pmax (MW) | Reactive Power Capability (MVar) | Proposed VAR Compensation | VAR Injection |          |                  | VAR Absorption |          |                   | Meet FERC Order 827 Requirement? | Add'l VAR Needed (MVar) |
|----------------|-----------|----------------------------------|---------------------------|---------------|----------|------------------|----------------|----------|-------------------|----------------------------------|-------------------------|
|                |           |                                  |                           | P (MW)        | Q (MVar) | Lagging p.f (pu) | P (MW)         | Q (MVar) | Leading p.f. (pu) |                                  |                         |
| J1205          | 125       | ±41.0900                         | 1 x 18 MVar Cap           | 123.4         | 43.5     | 0.9431           | 123.2          | -48.6    | -0.9302           | Yes                              |                         |
| J1215          | 155       | ±50.94                           | 2 x 3 MVar Cap            | 154           | 31.7     | 0.9795           | 153.7          | -70.3    | -0.9094           | No                               | 12.5                    |
| J1219          | 100       | ±32.87                           | 1 x 3 MVar Cap            | 99.1          | 15.6     | 0.9878           | 98.8           | -53.2    | -0.8805           | No                               | 13.8                    |
| J1239          | 100       | ±50                              | None                      | 99            | 39.2     | 0.9298           | 98.6           | -67      | -0.8271           | Yes                              |                         |
| J1257          | 20        | ±9.6900                          | None                      | 19.8          | 12.5     | 0.8456           | 19.8           | -14.7    | -0.8029           | Yes                              | 0.3                     |
| J1258          | 50        | ±16.4300                         | 1 x 2 MVar Cap            | 49.3          | 14.8     | 0.9578           | 49.2           | -22.5    | -0.9094           | No                               | 1.4                     |
| J1281          | 75        | ±50.2000                         | None                      | 74.9          | 37.9     | 0.8923           | 74.1           | -75.2    | -0.7019           | Yes                              |                         |
| J1286          | 125       | ±41.1000                         | 1 x 4 MVar Cap            | 123.2         | 45.6     | 0.9378           | 122.8          | -47.8    | -0.9319           | Yes                              |                         |
| J1294          | 99        | ±40.8000                         | None                      | 99            | 25.5     | 0.9684           | 97.1           | -67.1    | -0.8227           | No                               | 7                       |
| J1347          | 175       | ±101.3700                        | None                      | 172.8         | 83.2     | 0.901            | 169.9          | -155.4   | -0.7379           | Yes                              |                         |
| J1361          | 50        | ±17.3900                         | None                      | 52.1          | 8.3      | 0.9875           | 51.8           | -29.5    | -0.869            | No                               | 8.8                     |
| J1367          | 95        | ±32.63                           | None                      | 94.1          | 20.6     | 0.9769           | 93.8           | -47.9    | -0.8906           | No                               | 10.3                    |
| J1368          | 200       | ±65.7000                         | 1 x 10 MVar Cap           | 199.1         | 27.6     | 0.9905           | 197.2          | -142.6   | -0.8103           | No                               | 37.8                    |
| J1372          | 100       | ±38.5950                         | 1 x 12 MVar Cap           | 100.6         | 9.3      | 0.9958           | 98.2           | -62.1    | -0.8452           | No                               | 23.8                    |
| J1373          | 95        | ±45.6700                         | 1 x 6 MVar Cap            | 95            | 38.2     | 0.9278           | 94.4           | -73.2    | -0.7903           | Yes                              |                         |
| J1402 (w J919) | 180       | ±79.25                           | 1 x 42 MVar Cap           | 180           | 95.1     | 0.8842           | 179            | -131.7   | -0.8055           | Yes                              |                         |

| Project | Pmax (MW) | Reactive Power Capability (MVar) | Proposed VAR Compensation | VAR Injection |          |                  | VAR Absorption |          |                   | Meet FERC Order 827 Requirement? | Add'l VAR Needed (MVar) |
|---------|-----------|----------------------------------|---------------------------|---------------|----------|------------------|----------------|----------|-------------------|----------------------------------|-------------------------|
|         |           |                                  |                           | P (MW)        | Q (MVar) | Lagging p.f (pu) | P (MW)         | Q (MVar) | Leading p.f. (pu) |                                  |                         |
| J1415   | 250       | ±82.18                           | 2 x 4 MVar Cap            | 245.3         | 61.5     | 0.97             | 244.1          | -124.7   | -0.8905           | No                               | 19.1                    |
| J1421   | 300       | ±50.7400                         | 2 x 38 MVar Cap           | 295.2         | 144.4    | 0.8983           | 293.5          | -139.8   | -0.9028           | Yes                              |                         |
| J1424   | 200       | ±65.7400                         | 1 x 14 MVar Cap           | 197.8         | 48.2     | 0.9716           | 197.1          | -110.9   | -0.8715           | No                               | 18.7                    |
| J1431   | 90        | ±49.3000                         | None                      | 88.5          | 40.2     | 0.9105           | 88.2           | -50.2    | -0.8691           | Yes                              |                         |
| J1434   | 100       | ±32.87                           | 2 x 7 MVar                | 98.9          | 32.7     | 0.9494           | 98.6           | -53.9    | -0.8775           | Yes                              |                         |
| J1436   | 50        | ±16.53                           | 3 x 7.5                   | 49.9          | 37       | 0.8033           | 49.8           | -20.9    | -0.9221           | Yes                              |                         |
| J1437   | 180       | ±87.38                           | 4 x 10.5 MVar Cap         | 175.4         | 115.2    | 0.8358           | 151.5          | -162.4   | -0.6821           | Yes                              |                         |
| J1440   | 150       | ±49.303                          | 1 x 5 MVar Cap            | 148.9         | 32.3     | 0.9773           | 148.9          | -84.4    | -0.87             | No                               | 16.7                    |
| J1441   | 200       | ±71                              | 2 x 14 MVar Cap           | 200.1         | 66.7     | 0.9487           | 199.1          | -121.1   | -0.8544           | Yes                              |                         |
| J1442   | 105       | ±39.7400                         | 1 x 6 MVar Cap            | 104.5         | 36.4     | 0.9444           | 104.3          | -52.3    | -0.8939           | Yes                              |                         |
| J1455   | 150       | ±49.3                            | 2 x 11 MVar Cap           | 148.4         | 50.2     | 0.9473           | 147.8          | -82.1    | -0.8742           | Yes                              |                         |
| J1458   | 200       | ±65.74                           | 2 x 14 MVar Cap           | 197.4         | 63.7     | 0.9517           | 197            | -112.9   | -0.8676           | No                               | 1.2                     |
| J1463   | 150       | ±52.4                            | None                      | 147.4         | 23.5     | 0.9875           | 146.3          | -93.8    | -0.8418           | No                               | 24.9                    |
| J1465   | 150       | ±49.3                            | 2 x 14 MVar Cap           | 148.6         | 61.3     | 0.9244           | 148            | -76.2    | -0.8891           | Yes                              |                         |

### 3. Model Development and Study Assumptions

#### 3.1. Base Case Models

The origin of the DPP 2019 Cycle 1 South models is the MTEP19 models with the Bench Cases including all pre-queued projects and associated Network Upgrades, while the Study Cases contain all of the interconnection requests in DPP 2019 Cycle 1 South Phase III, in addition to all the facilities in the Bench Cases.

- Bench Cases
  - DPP19C1-Phase3-2024SUM-Bench
- Study Cases
  - DPP19C1-Phase3-2024SUM-Study-Discharging

#### 3.2. Monitored Elements

Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal rating (PSS®E Rating A), and for NERC category P1-P7 conditions branches were monitored for emergency rating (PSS®E Rating B). Voltage limits were specified for system intact and contingent conditions as per applicable Transmission Owner Planning Criteria.

#### 3.3. Contingencies

The following contingencies were considered in the steady state analysis:

- 1) NERC Category P0 (system intact -- no contingencies)

- 2) NERC Category P1 contingencies
  - a. Single element outages, at buses with a nominal voltage of 68 kV and above
  - b. Multiple element NERC Category P1 contingencies
- 3) NERC Category P2-P7 contingencies
- 4) For all the contingencies and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

### **3.4. Study Methodology**

Non-linear (AC) contingency analysis was performed on the benchmark and study cases, and the incremental impact of the DPP 2019 Cycle 1 South generating facilities was evaluated by comparing the steady state performance of the transmission system in the Bench and Study Cases. Analyses used PSS®E version 33.11.0 and TARA version 2001.

### **3.5. Performance Criteria**

A branch is considered a thermal constraint if the following conditions are met:

- 1) The generator has a larger than twenty percent (20%) sensitivity factor on the overloaded facilities under post-contingent condition (see NERC TPL) or five percent (5%) sensitivity factor under system-intact condition, or
- 2) The overloaded facility or the overload-causing contingency is at generator's outlet, or
- 3) The megawatt impact due to the generator is greater than or equal to twenty percent (20%) of the applicable rating (normal or emergency) of the overloaded facility, or
- 4) For any other constrained facility, where none of the Study Generators meet one of the above criteria, however, the cumulative MW impact of the group of study generators is greater than twenty percent (20%) of the rating of the facility, then only those study generators whose individual MW impact is greater than five percent (5%) of the rating of the facility and has DF greater than five percent (5%) will be responsible for mitigating the cumulative MW impact constraint, or
- 5) Impacts on Affected Systems would be classified as injection constraints based on the Affected Systems' criteria, or
- 6) Any other applicable Transmission Owner Local Planning Criteria are met.

A bus is considered a voltage constraint if both of the following conditions are met:

- 1) The bus voltage is outside of the applicable normal or emergency limits for the post change case, and
- 2) The change in bus voltage is greater than 0.01 per unit

All generators must mitigate thermal injection constraints and voltage constraints in order to obtain any type of Interconnection Service. Further, all generators requesting NRIS must mitigate constraints found by using the Deliverability algorithm to meet the system performance criteria for NERC category P1 events, if DFAX due to the study generator is equal to or greater than 5%.

## **4. Thermal Analysis**

The thermal analysis results for 2024 Summer show generator projects J1249 and J1421 causing constraints. The details pertaining to the thermal analysis can be found in Appendix C – MISO ERIS Analysis (CEII).

### **4.1. J1249**

Thermal analysis determined that this generator contributes to the following constraints:

1. Purvis – Oak Grove 69 kV
2. Purvis – Lumber 69 kV
3. Oak Grove 161/69 kV Transformer



Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraints. The planned upgrade is to add a new line from Morrow – Purvis and upgrade the Purvis substation. The cost estimate is \$6.772 million, with the project allocated 100% of the cost.

#### **4.2. J1421**

Thermal analysis determined that this generator contributes to the following constraints:

1. Coly – Moler 230 kV

Per MISO cost allocation rules, the project receives cost allocation for upgrades required to mitigate the listed constraint. The planned upgrade is to upgrade the Coly - Moler line. The cost estimate is \$19.6 million, with the project allocated 100% of the cost.

### **5. Voltage Analysis**

The voltage analysis results for 2024 Summer show no generator projects causing voltage violations. The details pertaining to the voltage analysis can be found in Appendix C – MISO ERIIS Analysis (CEII). Although a low voltage was initially identified on the Fourchon 115 kV bus, it was determined that changing the settings of existing equipment would be sufficient to mitigate the constraint and no network upgrades were required.

### **6. Stability Analysis**

Two stability analyses were performed on the 2019-Cycle 1-South-Phase III study units to determine their dynamics performance – one using the dispatch from the ERIIS discharging case and one capturing the Entergy Local Planning Criteria (LPC) for new generator interconnections. No issues were identified in the stability analysis.

The details pertaining to the stability analysis can be found in **Error! Reference source not found.** and **Error! Reference source not found.**

#### **6.1. Model Development**

The following summer peak 2024 models were developed based on Phase III study models. The Entergy LPC stability models were developed based on the Phase III stability study models and were adjusted in order to comply with Entergy's LPC by fully dispatching generation local to study projects.

- Bench Case
  - DPP19C1-Phase3-2024SUM-Bench
  - DPP19C1-Phase3-2024SUM-Bench\_Entergy LPC
- Study Case
  - DPP19C1-Phase3-2024SUM-Study
  - DPP19C1-Phase3-2024SUM-Study\_Entergy LPC

#### **6.2. Study Methodology**

The purpose of the study is to identify potential angular instabilities, voltage dip violations, and damping violations, if any, due to the interconnection of the projects in the DPP 2019-1 South study cycle under disturbance conditions, and the impact of all study projects on the system stability performance.

The fault scenarios simulated in this study cover faults simulated as part of the MTEP19 analysis as well as selected three-phase (3PH) faults with normal clearing and single line to ground (SLG) faults with delayed clearing. Dynamic simulations of fault scenarios were performed using the DSATools TSAT program (version 19.0.10).

Fault scenarios were first simulated using the study case and the results were reviewed. For scenarios that exhibited instability, the bench case was simulated such that the stability performance with and without the proposed interconnection projects could be compared. Any new stability problems attributed to the proposed

interconnection projects are flagged and reported.

For each fault, rotor angles, speed deviation, and electrical power outputs of the study generators and the generators in the proximity were monitored. Voltages at selected buses, including all POI buses of the study projects and all future buses, were also monitored.

In addition, Section 10 of the “Entergy Transmission Local Planning Criteria”<sup>1</sup> states the following:

- **New Generator Interconnections:** *For Generator Interconnection (GI) Stability Studies and TPL Reliability Stability Studies only, new generation requests (for GI studies) or generation with an executed Generator Interconnection Agreement (GIA) since performance of the last reliability assessment (for TPL studies) will be studied with all generation local to the new generator at full output to determine whether the aggregate of the generation can be injected into the transmission system without experiencing transient stability issues for local faults.*

For the purposes of the Entergy LPC study local generation has been defined as generation located within one breaker substation from the Point of Interconnection if interconnecting to an existing substation or located within one breaker substation from a termination substation when tapping a line.

The fault scenarios simulated in the Entergy LPC study cover selected 3PH and 3PH P3 contingency faults with normal clearing. In each fault scenario, a generator is disconnected at 0.5 seconds and the simulation continues to run until 5.0 seconds at which point the fault is initiated and the total simulation run time is 20.0 seconds.

### 6.3. Study Criteria

Transient stability study criteria are based upon Section 11.A of the “Entergy Transmission Local Planning Criteria”.

In addition, all renewable study projects are subject to the voltage ride-through and frequency ride-through criteria specified in NERC PRC-024-2 (“Generator Frequency and Voltage Protective Relay Settings”<sup>2</sup>) to check if the projects remain connected during frequency and voltage excursions. Specifically, PRC-024-2 mandates that protective relaying should be set in such a way that:

- Voltage Ride-Through: a generator shall withstand zero voltage at the POI (typically the primary side of the station transformer) for up to 0.15 seconds (9 cycles) and the ensuing voltage recovery period for three phase faults.
- Frequency Ride-Through: a generator shall maintain continuous operation between 59.5 and 60.5 Hz.

### 6.4. Study Results

The stability analyses showed that J1436 failed to recover to its nominal active power level after 3PH fault clearing. The complete list of 3PH and SLG faults simulated as well as their corresponding results and plots are included in **Error! Reference source not found.** and **Error! Reference source not found.**

## 7. Short Circuit Analysis

Short circuit analysis was performed utilizing ASPEN software. Single line to ground faults and three phase faults were evaluated for pre- and post-project cases (similar to Bench and Study models). The short circuit studies for the DPP 2019-1 South study cycle indicate that the study generators J1246, J1421, and J1463 have adverse impacts on circuit breaker capability that require upgrades. Cost estimates for the required upgrades will be provided in phase 3. The details pertaining to the short circuit analysis are presented in **Error! Reference source not found.**

<sup>1</sup> <https://cdn.misoenergy.org/Entergy%20TO%20Planning%20Criteria108226.pdf>

<sup>2</sup> <http://www.nerc.com/pa/Stand/Reliability%20Standards/PRC-024-2.pdf>

## **8. Affected System Impact Study**

The details pertaining to the AECI, SOCO, and TVA Affected Systems studies are in Appendix H – AECI Affected Systems Study Report (CEII), Appendix I – SOCO Affected Systems Study Report (CEII), and Appendix J – TVA Affected Systems Report (CEII). MLGW and PowerSouth did not study any projects in the DPP 2019 Cycle 1 South Phase III analysis. SPP did not find any impacts due to DPP 2019 Cycle 1 South units in Phase II, thus did not perform an analysis in Phase III. Projects not listed in this section did not contribute to any of the relevant constraints.

### **8.1. J1215**

The TVA Study did not identify any constraints associated with this generator.

### **8.2. J1249**

The SOCO Study identified the following costs for J1249.

#### **1. Carriere SW – Marion South 230 kV**

Per SOCO cost allocation rules, this project is assigned costs for the planned upgrade to build a new 230 kV 3 breaker ring switching station on Hattiesburg SW – Adams Creek 230 kV line, construct a new 230 kV 3 breaker ring bus at Carriere SW, and construct a new 33-mile line between Carriere SW and Marion S for \$52.7 million. This project is allocated 100% of the cost.

#### **2. Kiln – Bayou Lacroix 115 kV**

Per SOCO cost allocation rules, this project is assigned costs for the planned upgrade to reconductor/rebuild 6.2 miles of 115 kV line from Kiln – Bayou Lacroix for \$8.4 million. This project is allocated 100% of the cost.

#### **3. Hattiesburg Hwy 11 – Hattiesburg Co. Drive**

Per SOCO cost allocation rules, this project is assigned costs for the planned upgrade to rebuild 3.2 miles from Hattiesburg Hwy 11 – Hattiesburg Co Drive for \$2.2 million. This project is allocated 100% of the cost.

### **8.3. J1294**

The TVA Study did not identify any constraints associated with this generator.

### **8.4. J1347**

The SOCO Study identified the following costs for J1347.

#### **1. Forest Industrial – Newton 115 kV**

Per SOCO cost allocation rules, this project is assigned costs for the planned upgrade to reconductor/rebuild 23 miles of Forest Industrial – Newton line for \$17 million. This project is allocated 100% of the cost.

### **8.5. J1434**

The TVA Study did not identify any constraints associated with this generator.

### **8.6. J1437**

The AECI Study identified that this generator contributes to the following constraints:

1. Axtell - Macon Lake 69 kV
2. Axtell - Macon Tap 69 kV
3. Bevier - Bevier Tap 69 kV
4. Elm - Holden 69 kV
5. Gobknob - PB South 69 kV
6. Holden 161/69 kV transformer



7. Love Lake - Macon Tap 69 kV
8. Macon Lake - Bevier Tap 69 kV
9. Thomas Hill - Adair 161 kV

Per cost allocation rules, the project is assigned costs for upgrades required to mitigate the listed constraints. This project's costs are approximately \$1,150,239.

#### 8.7. J1440

The TVA Study did not identify any constraints associated with this generator.

#### 8.8. J1458

The TVA Study did not identify any constraints associated with this generator.

### 9. Deliverability Analysis

Generator interconnection projects must pass Generator Deliverability Study to be granted NRIS. If the generator is deemed not fully deliverable, the customer can choose either to change the project to an Energy Resource project or to proceed with the system upgrades that will make the generator fully deliverable. Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottled up. The wind and solar generators are tested at 100% of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TEMT).

The MISO Generator Deliverability Study whitepaper describing the algorithm can be found in BPM 015 – Generation Interconnection, Appendix C.

#### 9.1. Determining the MW Restriction

If one facility is overloaded based on the assessed “severe yet credible dispatch” scenario described in the study methodology, and the generator under study has a DF greater than 5%, part or all of its output it is not deliverable. The restricted MW is calculated as following:

$$(\text{MW restricted}) = (\text{worst loading} - \text{MW rating}) / (\text{generator sensitivity factor})$$

If the result is larger than the maximum output of the generator, 100% of this generator's output is not deliverable.

#### 9.2. Deliverability Study Results

The limiting constraints (monitored facilities – contingency facility pairs) seen in the deliverability analysis for the 2024 Summer case are summarized in Appendix D – MISO Deliverability Analysis (CEII).

##### 9.2.1 J1215

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 5 shows the NRIS results and cost estimates determined in the NRIS analysis.

**Table 5: NRIS Results for J1215**

| J1215 Deliverable (NRIS) Amount in 2024 Case:<br>(Conditional on ERI and case assumptions)           |                             |                     | 0 MW (0%)                                |                                     |                            |
|------------------------------------------------------------------------------------------------------|-----------------------------|---------------------|------------------------------------------|-------------------------------------|----------------------------|
| Next Upgrade for Higher NRIS Level<br>(cumulative)<br>(i.e. All upgrades must be made for 100% NRIS) | Level of Service Attainable | Distribution Factor | Projects Associated With NRIS Constraint | NRIS Cost Allocated to Project (\$) | Total Cost of Upgrade (\$) |

|                                                    | ble<br>(MW) |       |                                                       |            |             |
|----------------------------------------------------|-------------|-------|-------------------------------------------------------|------------|-------------|
| 336839 8R.BRASWELL% 500 336562<br>8FRANKLIN% 500 1 | 155         | 0.059 | J1425, J1434, J1215,<br>J1440, J1281, J1294,<br>J1458 | \$ 341,594 | \$3,000,000 |

### 9.2.2 J1219

This generator is determined to be fully deliverable to 100 MW.

### 9.2.3 J1246

This generator is determined to be deliverable up to 439.41 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 6 shows the NRIS results and cost estimates determined in the NRIS analysis.

**Table 6: NRIS Results for J1246**

| J1246 Deliverable (NRIS) Amount in 2024 Case:<br>(Conditional on ERIS and case assumptions)          |                                  | 439.41 MW (62.3%)       |                                          |                                     |                            |
|------------------------------------------------------------------------------------------------------|----------------------------------|-------------------------|------------------------------------------|-------------------------------------|----------------------------|
| Next Upgrade for Higher NRIS Level<br>(cumulative)<br>(i.e. All upgrades must be made for 100% NRIS) | Level of Service Attainable (MW) | Distributio<br>n Factor | Projects Associated With NRIS Constraint | NRIS Cost Allocated to Project (\$) | Total Cost of Upgrade (\$) |
| 336010 8BYULABUTTE% 500 336011<br>6BYULABUTTE! 230 1                                                 | 563.54                           | 0.2905                  | J1246                                    | \$42,256,011                        | \$42,256,011               |
| 336010 8BYULABUTTE% 500 335618<br>8WILLOW_GL% 500 1                                                  | 705                              | 0.6323                  | J1455, J1367, J1465,<br>J1246            | \$18,228,105                        | \$20,243,989               |

### 9.2.4 J1249

This generator is determined to be deliverable up to 532.49 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 7 shows the NRIS results and cost estimates determined in the NRIS analysis.

**Table 7: NRIS Results for J1249**

| J1249 Deliverable (NRIS) Amount in 2024 Case:<br>(Conditional on ERIS and case assumptions)          |                                  | 532.49 MW (88.7%)       |                                          |                                     |                            |
|------------------------------------------------------------------------------------------------------|----------------------------------|-------------------------|------------------------------------------|-------------------------------------|----------------------------|
| Next Upgrade for Higher NRIS Level<br>(cumulative)<br>(i.e. All upgrades must be made for 100% NRIS) | Level of Service Attainable (MW) | Distrib<br>ution Factor | Projects Associated With NRIS Constraint | NRIS Cost Allocated to Project (\$) | Total Cost of Upgrade (\$) |
| 318004 5PURVIS 161 388310<br>6PURVIS B1 230 1                                                        | 534.4                            | 0.5058                  | J1249                                    | \$ -                                | \$ -                       |
| 318004 5PURVIS 161 388301<br>6PURVIS B2 230 1                                                        | 600                              | 0.5043                  | J1249                                    | \$ -                                | \$ -                       |

### 9.2.5 J1257

This generator is determined to be fully deliverable to 20 MW.

### 9.2.6 J1258

This generator is determined to be fully deliverable to 50 MW.

### 9.2.7 J1281

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 8 shows the NRIS results and cost estimates determined in the NRIS analysis.

**Table 8: NRIS Results for J1281**

| J1281 Deliverable (NRIS) Amount in 2024 Case:<br>(Conditional on ERIS and case assumptions)       |                                  | 0 MW (0%)           |                                                          |                                     |                            |
|---------------------------------------------------------------------------------------------------|----------------------------------|---------------------|----------------------------------------------------------|-------------------------------------|----------------------------|
| Next Upgrade for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS) | Level of Service Attainable (MW) | Distribution Factor | Projects Associated With NRIS Constraint                 | NRIS Cost Allocated to Project (\$) | Total Cost of Upgrade (\$) |
| 336839 8R.BRASWELL% 500<br>336562 8FRANKLIN% 500 1                                                | 0                                | 0.0882              | J1425, J1434,<br>J1215, J1440,<br>J1281, J1294,<br>J1458 | \$ 247,091                          | \$ 3,000,000               |
| 336129 6BOGALUSA 230 336131<br>6ADMSCRK% 230 Z1                                                   | 45.98                            | 0.0628              | J1281, J1294,<br>J1458                                   | \$ 2,495,091                        | \$ 15,000,000              |
| 336130 8BOGALUS 500 336129<br>6BOGALUSA 230 1                                                     | 74.81                            | 0.0628              | J1281, J1294,<br>J1458                                   | \$ -                                | \$ -                       |
| 337040 6ANDRUS% 230 998764<br>6ANDRUS% 115 1                                                      | 75                               | 0.0855              | J1281, J1458                                             | \$ -                                | \$ -                       |

### 9.2.8 J1294

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 9 shows the NRIS results and cost estimates determined in the NRIS analysis.

**Table 9: NRIS Results for J1294**

| J1294 Deliverable (NRIS) Amount in 2024 Case:<br>(Conditional on ERIS and case assumptions)       |                                  | 0 MW (0%) |                                                    |                                     |                            |
|---------------------------------------------------------------------------------------------------|----------------------------------|-----------|----------------------------------------------------|-------------------------------------|----------------------------|
| Next Upgrade for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS) | Level of Service Attainable (MW) | DFAX      | Projects Associated with NRIS Constraint           | NRIS Cost Allocated to Project (\$) | Total Cost of Upgrade (\$) |
| 336839 8R.BRASWELL% 500<br>336562 8FRANKLIN% 500 1                                                | 0                                | 0.0882    | J1425, J1434, J1215, J1440, J1281,<br>J1294, J1458 | \$ 247,091                          | \$ 3,000,000               |
| 336129 6BOGALUSA 230 336131<br>6ADMSCRK% 230 Z1                                                   | 45.98                            | 0.0628    | J1281, J1294, J1458                                | \$ 4,111,656                        | \$ 15,000,000              |
| 336130 8BOGALUS 500 336129<br>6BOGALUSA 230 1                                                     | 74.81                            | 0.0628    | J1281, J1294, J1458                                | \$ -                                | \$ -                       |
| 336994 3ISOLA+ 115 336993<br>3BELZONI 115 1                                                       | 75                               | 0.0855    | J1281, J1458                                       | \$ 250,314                          | \$ -                       |

### 9.2.9 J1361

This generator is determined to be fully deliverable to 50 MW.

### 9.2.10 J1367

This generator is determined to be deliverable up to 38.89 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 10 shows the NRIS results and cost estimates determined in the NRIS analysis.

**Table 10: NRIS Results for J1367**

| J1367 Deliverable (NRIS) Amount in 2024 Case:<br>(Conditional on ERS and case assumptions)        |                                  | 38.89 MW (80%) |                                          |                                     |                            |
|---------------------------------------------------------------------------------------------------|----------------------------------|----------------|------------------------------------------|-------------------------------------|----------------------------|
| Next Upgrade for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS) | Level of Service Attainable (MW) | DFAX           | Projects Associated with NRIS Constraint | NRIS Cost Allocated to Project (\$) | Total Cost of Upgrade (\$) |
| 336010 8BYULABUTTE% 500<br>335618 8WILLOW_GL% 500 1                                               | 48.65                            | 0.1935         | J1455, J1367, J1465, J1246               | \$435,444                           | \$20,243,989               |

### 9.2.11 J1368

This generator is determined to be fully deliverable up to 40 MW.

### 9.2.12 J1372

This generator is determined to be fully deliverable up to 20 MW.

### 9.2.13 J1373

This generator is determined to be fully deliverable up to 80.75 MW.

### 9.2.14 J1415

This generator is determined to be fully deliverable to 250 MW.

### 9.2.15 J1421

This generator is determined to be fully deliverable to 300 MW.

### 9.2.16 J1425

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 11 shows the NRIS results and cost estimates determined in the NRIS analysis.

**Table 11: NRIS Results for J1425**

| J1421 Deliverable (NRIS) Amount in 2024 Case:<br>(Conditional on ERS and case assumptions)        |                                  | 0 MW (0%) |                                                 |                                     |                            |
|---------------------------------------------------------------------------------------------------|----------------------------------|-----------|-------------------------------------------------|-------------------------------------|----------------------------|
| Next Upgrade for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS) | Level of Service Attainable (MW) | DFAX      | Projects Associated with NRIS Constraint        | NRIS Cost Allocated to Project (\$) | Total Cost of Upgrade (\$) |
| 336839 8R.BRASWELL% 500<br>336562 8FRANKLIN% 500 1                                                | 144                              | 0.0546    | J1425, J1434, J1215, J1440, J1281, J1294, J1458 | \$ 232,501                          | \$3,000,000                |

### 9.2.17 J1431

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 12 shows the NRIS results and cost estimates determined in the NRIS analysis.

**Table 12: NRIS Results for J1431**

| J1431 Deliverable (NRIS) Amount in 2024 Case:<br>(Conditional on ERI and case assumptions)              |                                           | 0 MW (0%) |                                             |                                           |                               |  |
|---------------------------------------------------------------------------------------------------------|-------------------------------------------|-----------|---------------------------------------------|-------------------------------------------|-------------------------------|--|
| Next Upgrade for Higher NRIS<br>Level (cumulative)<br>(i.e. All upgrades must be made for<br>100% NRIS) | Level of<br>Service<br>Attainable<br>(MW) | DFAX      | Projects Associated with NRIS<br>Constraint | NRIS Cost<br>Allocated to Project<br>(\$) | Total Cost of<br>Upgrade (\$) |  |
| 336190 6GYPSY% 230 336189<br>6IRONMAN% 230 1                                                            | 15.79                                     | 0.0509    | J1441, J1431                                | \$ 151,478                                | \$ 500,000                    |  |
| 336066 6FRISCO! 230 336065<br>6CONVNT! 230 1                                                            | 90                                        | 0.0842    | J1441, J1431                                | \$ 7,326,200                              | \$ 23,800,000                 |  |

### 9.2.18 J1434

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 13 shows the NRIS results and cost estimates determined in the NRIS analysis.

**Table 13: NRIS Results for J1434**

| J1434 Deliverable (NRIS) Amount in 2024 Case:<br>(Conditional on ERI and case assumptions)              |                                           | 0 MW (0%) |                                                    |                                        |                               |  |
|---------------------------------------------------------------------------------------------------------|-------------------------------------------|-----------|----------------------------------------------------|----------------------------------------|-------------------------------|--|
| Next Upgrade for Higher NRIS<br>Level (cumulative)<br>(i.e. All upgrades must be made for<br>100% NRIS) | Level of<br>Service<br>Attainable<br>(MW) | DFAX      | Projects Associated with NRIS<br>Constraint        | NRIS Cost Allocated<br>to Project (\$) | Total Cost of<br>Upgrade (\$) |  |
| 336839 8R.BRASWELL% 500<br>336562 8FRANKLIN% 500 1                                                      | 100                                       | 0.0609    | J1425, J1434, J1215, J1440, J1281,<br>J1294, J1458 | \$ 227,480                             | \$ 3,000,000                  |  |

### 9.2.19 J1436

This generator is determined to be fully deliverable to 50 MW.

### 9.2.20 J1437

This generator is determined to be fully deliverable to 180 MW.

### 9.2.21 J1440

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 14 shows the NRIS results and cost estimates determined in the NRIS analysis.

**Table 14: NRIS Results for J1440**

| J1440 Deliverable (NRIS) Amount in 2024 Case:<br>(Conditional on ERI and case assumptions) |  | 0 MW (0%) |  |  |  |  |
|--------------------------------------------------------------------------------------------|--|-----------|--|--|--|--|
|--------------------------------------------------------------------------------------------|--|-----------|--|--|--|--|

| Next Upgrade for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS) | Level of Service Attainable (MW) | DFAX   | Projects Associated with NRIS Constraint        | NRIS Cost Allocated to Project (\$) | Total Cost of Upgrade (\$) |
|---------------------------------------------------------------------------------------------------|----------------------------------|--------|-------------------------------------------------|-------------------------------------|----------------------------|
| 336839 8R.BRASWELL% 500<br>336562 8FRANKLIN% 500 1                                                | 150                              | 0.0727 | J1425, J1434, J1215, J1440, J1281, J1294, J1458 | \$ 407,336                          | \$ 3,000,000               |

#### 9.2.22 J1441

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 15 shows the NRIS results and cost estimates determined in the NRIS analysis.

**Table 15: NRIS Results for J1441**

| J1441 Deliverable (NRIS) Amount in 2024 Case:<br>(Conditional on ERIS and case assumptions)       |                                  |        | 0 MW (0%)                                |                                     |                            |  |
|---------------------------------------------------------------------------------------------------|----------------------------------|--------|------------------------------------------|-------------------------------------|----------------------------|--|
| Next Upgrade for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS) | Level of Service Attainable (MW) | DFAX   | Projects Associated with NRIS Constraint | NRIS Cost Allocated to Project (\$) | Total Cost of Upgrade (\$) |  |
| 336190 6GYPSY% 230 336189<br>6IRONMAN% 230 1                                                      | 35.08                            | 0.0527 | J1441, J1431                             | \$ 348,522                          | \$ 500,000                 |  |
| 336066 6FRISCO! 230 336065<br>6CONVNT! 230 1                                                      | 200                              | 0.0852 | J1441, J1431                             | \$ 16,473,800                       | \$ 23,800,000              |  |

#### 9.2.23 J1455

This generator is determined to be deliverable up to 119.91 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 16 shows the NRIS results and cost estimates determined in the NRIS analysis.

**Table 16: NRIS Results for J1455**

| J1455 Deliverable (NRIS) Amount in 2024 Case:<br>(Conditional on ERIS and case assumptions)       |                                  |        | 76.82 MW (51.12%)                        |                                     |                            |  |
|---------------------------------------------------------------------------------------------------|----------------------------------|--------|------------------------------------------|-------------------------------------|----------------------------|--|
| Next Upgrade for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS) | Level of Service Attainable (MW) | DFAX   | Projects Associated with NRIS Constraint | NRIS Cost Allocated to Project (\$) | Total Cost of Upgrade (\$) |  |
| 336010 8BYULABUTTE% 500<br>335618 8WILLOW_GL% 500 1                                               | 150                              | 0.2067 | J1455, J1367, J1465, J1246               | \$1,434,169                         | \$20,243,989               |  |

#### 9.2.24 J1458

This generator is determined to be deliverable up to 0 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 17 shows the NRIS results and cost estimates determined in the NRIS analysis.

**Table 17: NRIS Results for J1458**

| J1458 Deliverable (NRIS) Amount in 2024 Case:<br>(Conditional on ERI and case assumptions)        |                                  |        | 0 MW (0%)                                       |                                     |                            |
|---------------------------------------------------------------------------------------------------|----------------------------------|--------|-------------------------------------------------|-------------------------------------|----------------------------|
| Next Upgrade for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS) | Level of Service Attainable (MW) | DFAX   | Projects Associated with NRIS Constraint        | NRIS Cost Allocated to Project (\$) | Total Cost of Upgrade (\$) |
| 336839 8R.BRASWELL% 500<br>336562 8FRANKLIN% 500 1                                                | 0                                | 0.1629 | J1425, J1434, J1215, J1440, J1281, J1294, J1458 | \$ 1,034,420                        | \$ 3,000,000               |
| 336129 6BOGALUSA 230 336131<br>6ADMSCRK% 230 Z1                                                   | 104.21                           | 0.0932 | J1281, J1294, J1458                             | \$ 8,393,253                        | \$ 15,000,000              |
| 336130 8BOGALUS 500 336129<br>6BOGALUSA 230 1                                                     | 119.74                           | 0.0932 | J1281, J1294, J1458                             | \$ -                                | \$ -                       |
| 337044 3GREENVILLE! 115<br>337045 3SE.GREENVL 115 1                                               | 142.48                           | 0.0681 | J1458                                           | \$ 5,300,000                        | \$ 5,300,000               |
| 336994 3ISOLA+ 115 336993<br>3BELZONI 115 1                                                       | 149.07                           | 0.0535 | J1294, J1458                                    | \$ 249,686                          | \$ 500,000                 |
| 337042 3ANDRUS! 115 337068<br>3REFUGE+ 115 1                                                      | 169.57                           | 0.1246 | J1458                                           | \$ -                                | \$ -                       |
| 337040 6ANDRUS% 230 998764<br>6ANDRUS% 115 1                                                      | 170                              | 0.2479 | J1281, J1458                                    | \$ -                                | \$ -                       |

### 9.2.25 J1465

This generator is determined to be deliverable up to 81.19 MW. Required upgrades to attain higher deliverable levels were identified in the NRIS analysis. Table 18 shows the NRIS results and cost estimates determined in the NRIS analysis.

**Table 18: NRIS Results for J1465**

| J1465 Deliverable (NRIS) Amount in 2024 Case:<br>(Conditional on ERI and case assumptions)        |                                  |        | 81.19 MW (54.13%)                        |                                     |                            |
|---------------------------------------------------------------------------------------------------|----------------------------------|--------|------------------------------------------|-------------------------------------|----------------------------|
| Next Upgrade for Higher NRIS Level (cumulative)<br>(i.e. All upgrades must be made for 100% NRIS) | Level of Service Attainable (MW) | DFAX   | Projects Associated with NRIS Constraint | NRIS Cost Allocated to Project (\$) | Total Cost of Upgrade (\$) |
| 336010 8BYULABUTTE% 500<br>335618 8WILLOW_GL% 500 1                                               | 150                              | 0.0592 | J1455, J1367, J1465, J1246               | \$410,754                           | \$20,243,989               |

## 10. Shared Network Upgrades Analysis

Shared Network Upgrade (SNU) Analysis, which tests for Network Upgrades driven by higher queued interconnection projects, was performed for this System Impact Study. No Shared Network Upgrades were

identified for any of the 2019-Cycle 1-South-Phase III projects.

## 11. Cost Allocation

The cost allocation of Network Upgrades (NU) for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of NRIS as of the draft System Impact Study report date.

### 11.1 Cost Assumptions for Network Upgrades

The cost estimate for each NU identified in the System Impact Study is provided by the corresponding Transmission Owner.

### 11.2 Cost Allocation Methodology

The costs of NU for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Study Case.

#### Cost Allocation Methodology for Thermal Constraints

1. With all study group generation projects dispatched in the Study Case, all thermal constraints are identified.
2. Distribution factor from each project on each constraint is obtained.
3. For each thermal constraint, the maximum MW contribution (increasing flow) from each project is then calculated in the Post Case without any Network Upgrades.
4. For each thermal constraint, the cost estimates for one or a subset of NU are provided by the corresponding Transmission Owner.
5. The cost of each NU is allocated based on the pro rata share of the MW contribution from each project on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of one NU is:

$$Project\ A\ Cost\ Portion\ of\ NU = \frac{Max(Proj.\ A\ MW\ contribution\ on\ constraint)}{\sum_i Max(Proj.\ i\ MW\ contrution\ on\ constraint)}$$

6. The total NU costs for each project are calculated if more than one NU is required.



**Appendix A – Cost Allocation Summary (CEII)**

**Appendix B – Network Upgrade Table Per Project (CEII)**

**Appendix C – MISO ERIS Analysis (CEII)**

**Appendix D – MISO Deliverability Analysis (CEII)**

**Appendix E – MISO Stability Plots (CEII)**

**Appendix F – Entergy LPC Stability Plots (CEII)**

**Appendix G – Short Circuit Study**

**Appendix H – AECI Affected Systems Report**

**Appendix I – SOCO Affected Systems Report**

**Appendix J – TVA Affected Systems Report**

In their report, TVA has requested that MISO coordinate with Memphis Light, Gas and Water (MLGW) to produce a new affected systems study to identify impacts on their system. However, all potentially affected systems were given an opportunity to indicate that they would require a study prior to and during the DPP-2019-1 South Phase 2 kick off. Since neither TVA nor MLGW indicated that a study on MLGW's system was required during that opportunity, MISO will not be considering any additional affected system studies for DPP-2019-1.

**Appendix K – A10 Results**